# NBA Shot Prediction

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### Abstract

*This project aims to predict NBA shot outcomes (made or missed) using a comprehensive dataset of NBA shots and player biometric information from the 2004-2024 seasons. The methodology involved retrieving and merging data from Kaggle, extensive exploratory data analysis, and robust feature engineering, including the removal of multicollinear and redundant features, followed by one-hot encoding. Multiple machine learning models were trained and evaluated, including Logistic Regression as a baseline, Random Forest, XGBoost, a scikit-learn Multi-Layer Perceptron (MLP), and a PyTorch Neural Network optimized using Optuna for hyperparameter tuning. All developed models demonstrated improved performance over the baseline, with the PyTorch Neural Network achieving the highest F1-score on the test set. Furthermore, a temporal analysis was conducted using LightGBM classifiers trained on rolling and expanding time windows, revealing a slight recency bias where models trained on newer data performed marginally better, reflecting the evolving dynamics of NBA basketball. This indicates that while current models offer reasonable predictive capabilities, continuous data updates may be beneficial for maintaining optimal performance.*

## I. Introduction

Basketball is a sport rich in data, with every action on the court meticulously recorded. Understanding the factors that contribute to a successful shot is a critical aspect for coaches, analysts, and players alike, influencing strategic decisions, player development, and game outcomes. The motivation behind this project is to leverage this data to gain insights into shot predictability and the evolving dynamics of the game.

This project addresses the problem of accurately predicting whether an NBA shot will be made or missed. Given the fast-paced and dynamic nature of basketball, even marginal improvements in shot prediction can yield significant analytical advantages. We aim to build robust predictive models that can identify key determinants of shot success, considering both contextual game-time factors and inherent player attributes.

The primary objectives of this project are twofold: first, to develop and evaluate various machine learning and deep learning models for binary shot outcome prediction; and second, to investigate the temporal stability and relevance of these models across different NBA seasons. The scope of this analysis spans from the 2004 to 2024 NBA seasons, utilizing a comprehensive dataset that combines shot-by-shot data with player biometric information.

Our methodology involves several key stages: initial data retrieval from public Kaggle repositories, merging disparate datasets (NBA shot data and player biometric data), followed by an extensive exploratory data analysis to understand data distributions and relationships. Feature engineering focuses on creating relevant predictors and addressing issues like multicollinearity and data leakage by dropping redundant spatial features, rare action types, and sensitive DRAFT_YEAR and EVENT_TYPE columns. Data is then preprocessed using one-hot encoding and standardized for model compatibility. We employ a range of models, including Logistic Regression, Random Forest, XGBoost, scikit-learn's Multi-Layer Perceptron (MLP), and a custom PyTorch Neural Network, with hyperparameters tuned using RandomizedSearchCV and Optuna. Finally, a temporal analysis using LightGBM is performed to assess how model performance changes when trained on different historical windows of data, providing insights into the concept of 'recency bias' in basketball analytics.

## II. Related Work

Shot outcome prediction (made or missed) is a foundational task in sports analytics, with roots in early machine learning applications to NBA play-by-play and shot-log data. Foundational studies established the predictive value of spatial features such as shot distance, location coordinates, and shot type. Meehan [1] applied logistic regression, random forests, and XGBoost to the 2014-15 Kaggle shot dataset, achieving approximately 68% accuracy with a tuned XGBoost model and identifying distance from the basket as the dominant predictor. Similarly, Oughali et al. [2] compared random forest and XGBoost classifiers on player shot data and reported comparable performance, confirming that tree-based ensembles effectively capture non-linear spatial relationships. More recent work by Alves et al. [6] extended these efforts to deep neural networks and gradient boosting variants, consistently observing test accuracies in the 64-67% range and noting the inherent stochastic ceiling of basketball shooting.

Subsequent research incorporated richer player-tracking data (e.g., SportVU) to model defender proximity, shot arc, and defensive contest intensity. Harmon et al. [3] leveraged multi-agent adversarial trajectories with convolutional and feed-forward networks, reaching roughly 61% accuracy and demonstrating the benefit of dynamic spatial context. Sliz [4] focused specifically on three-point shooting and highlighted defender distance and ball movement as critical variables.

A parallel line of inquiry has examined long-term temporal trends in NBA shooting. Zajac et al. [5] analyzed 40 consecutive seasons and documented the structural impact of the "three-point revolution," showing marked increases in three-point volume and efficiency alongside shifts in overall shot distribution. These temporal studies underscore that shooting dynamics are not stationary, yet few prior works systematically evaluate model stability across two decades using rolling or expanding training windows.

While these contributions provide strong baselines, several important gaps remain. First, the majority of existing models rely exclusively on shot-context features (location, time remaining, action type) or tracking-derived variables; very few integrate generalized player biometric attributes such as height, weight, and age, which may interact meaningfully with optimal shooting zones and fatigue effects. Second, temporal analyses are often limited to cross-sectional comparisons or short windows, leaving open questions about recency bias and the relevance of historical data amid league-wide strategic shifts. Third, although ensemble methods and basic neural networks are well explored, comprehensive benchmarking that includes hyperparameter-optimized deep networks (e.g., PyTorch architectures tuned via Optuna) alongside dedicated temporal-stability experiments is rare. Finally, many studies stop at performance reporting without deeply interpreting the persistent accuracy ceiling or linking it to unobservable factors such as defender skill and momentary physical nuances.

This project builds directly upon the spatial-modeling and ensemble foundations established by Meehan [1], Oughali et al. [2], and Harmon et al. [3] while addressing the identified gaps through two key innovations. We construct a unified dataset by merging comprehensive Kaggle shot logs (2004-2024) with a custom biometric dataset derived via the nba_api (height, weight, age, draft year), enabling the first large-scale examination of how player physical attributes moderate shot success. Furthermore, we extend prior temporal work (Zajac et al. [5]) by conducting an explicit LightGBM-based rolling- and expanding-window analysis over the full 20-season span, quantifying recency bias and model degradation across the three-point revolution. By evaluating a full spectrum of models—from logistic regression baselines to XGBoost, scikit-learn MLPs, and a custom Optuna-optimized PyTorch neural network—we provide both stronger predictive performance and clearer insight into the practical limits of shot predictability. These extensions differentiate our work from prior research and contribute novel, reproducible evidence to the sports analytics literature.

## III. Methodology

This project employs a systematic machine learning pipeline to predict NBA shot outcomes (either made or missed) by integrating shot spatial and contextual data with biometric attributes across the 2004-2024 NBA seasons. The methodology consists of data acquisition and integration, preprocessing and feature engineering, and model development and evaluation. This model development is further split into a general shot prediction where we try to maximize performance and evaluation of temporal stability and relevance of these models across different NBA seasons.

The dataset for this project was constructed by merging publicly available Kaggle datasets. First we have a collection of NBA shot datasets spanning the 2004-2024 seasons. Then, we have player biometric datasets spanning each year in the NBA containing physical attributes of the player such as height, weight, and age. These datasets are joined on player identities and season information in order to produce a singular unified dataset that captures both contextual shot information and generalized player characteristics.

After joining the datasets, player identifying information was dropped in the best interest of making the model more generalizable. We aimed to do this in order to test if the purely biometric and spatial information of a basketball shot could be used to accurately predict the result of a basketball shot. Further extending this analysis, we aimed to use these features to test if the fundamental deciders or factors used to predict basketball shots have changed across eras (such as with the recent three point revolution).

With our full datasets compiled, extensive exploratory data analysis (EDA) was conducted to understand feature distributions, detect anomalies, and identify relationships, with particular attention given to spatial (e.g., shot distance, court coordinates) and temporal (e.g., game clock, period) features hypothesized to have strong predictive power. Building on this analysis, critical feature engineering was applied to ensure model robustness by removing redundant, highly correlated features to reduce multicollinearity. Additionally, categorical variables such as shot type, action type, and team information were one-hot encoded, while identifiers and potentially leakage-prone attributes (e.g., event-specific metadata) were explicitly excluded to prevent the models from learning spurious patterns.

Our modeling approach incorporated a diverse set of machine learning algorithms in order to capture both linear and non-linear relationships in the data. A logistic regression model was implemented as a baseline due to its interpretability. More complex ensemble models, specifically Random Forest and XGBoost, were used to capture complex feature interactions. Neural Network approaches were also explored, including a MLP implemented in scikit-learn and a custom feedforward neural network built in PyTorch.

Hyperparameter optimization was performed using RandomizedSearchCV for traditional models and Optuna was used for the PyTorch neural network. These optimization techniques were used to systematically explore the parameter spaces to identify configurations that would maximize validation performance.

For our temporal modeling component, we introduced LightGBM regressors trained on a variety of different scenarios. LightGBM was selected due to being fast and robust allowing for quick experimentation. Scenarios for testing include training on rolling and expanding time windows, alongside having a head to head comparison of models trained on the

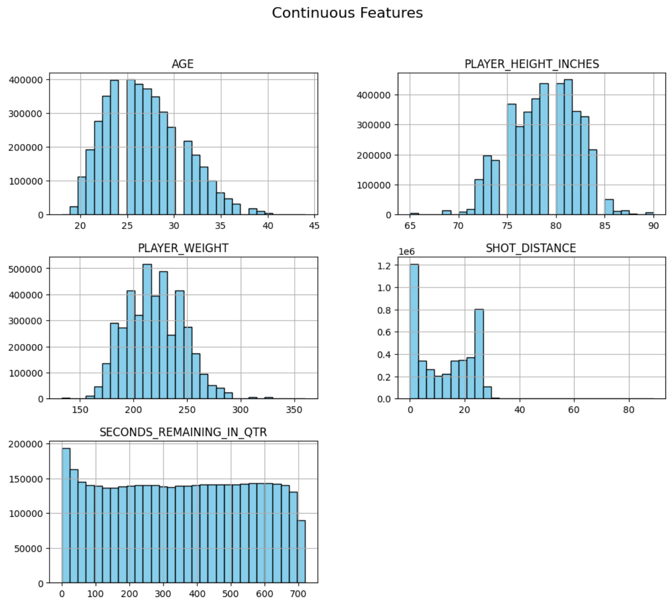


Fig. 1: Exploratory Data Analysis for Continuous Variables

oldest and newest data. These tests allowed for the evaluation of how model performance changes when trained on different historical subsets of data, providing insight into the impact of temporal drift and evolving game strategies.

Our methodology was motivated by the need to balance interpretability, predictive power, and generalization. By combining linear baselines, ensemble methods, and neural networks, our approach ensures comprehensive coverage of different modeling assumptions while enabling meaningful performance comparisons.

## IV. Experimental Setup

To rigorously evaluate the proposed shot-outcome prediction models, we conducted a comprehensive set of experiments on the unified dataset spanning the 2004-2024 NVA seasons. All experiments were implemented in Python using scikit-learn, XGBoost, LightGBM, and PyTorch libraries. These were executed in a Google Colab environment with GPU acceleration utilized for neural network training.

The dataset that we used for testing was created by merging the Kaggle NBA Shots dataset [7] with the custom biometric dataset [8] on PLAYER_ID and season information. No preexisting dataset matched our requirements from our methodology, so we generated 20 unique biometric datasets corresponding to each NBA season using the NBA API. The data pulled from the API contained one row per player per season with only the fields necessary for downstream joins with shot data. In total, over 40 datasets were merged and anonymized into a single unified dataset, with irrelevant or redundant columns removed to streamline the feature space. After consolidation, the final dataset was published to Kaggle so our main Colab notebook could load it directly without repeated API calls.

After consolidating our dataset, we first split into training, validation, and test sets. A standard train-test split was applied, with a portion of the training data further reserved for validation during hyperparameter tuning. Care was taken to ensure that preprocessing steps such as scaling and encoding were applied consistently across all splits to prevent data leakage.

To ensure data quality and prevent leakage, several preprocessing techniques were implemented. Redundant and multicollinear spatial features were removed, along with rare action types and the sensitive DRAFT_YEAR column. The target derived EVENT_TYPE column was dropped to preserve prediction integrity. Categorical variables were transformed via one-hot encoding in order to ensure suitable representations for both tree based and neural modals. Simultaneously, numerical features were standardized using z-score normalization to ensure compatibility with models sensitive to feature scaling, such as Logistic Regression. This process gave us our final feature set consisting of spatial coordinates, game context variables, and player biometrics.

To address class imbalances between made and missed shots, class weights were incorporated into certain models, such as Logistic Regression, to prevent bias toward the majority class. We evaluated multiple metrics for each model, including accuracy, precision, recall, and F1-score. We particularly placed an emphasis on F1-score during model evaluation due to the ability to balance precision and recall in the binary classification setting.

Each model was trained under controlled configurations. Logistic Regression was implemented with balanced class weights and regularization tuning. Random Forest and XGBoost models were configured with varying numbers of estimators, depth limits, and learning rates. Optimal parameters for these models were selected via RandomSearchCV with 8 trials and 3-fold cross validation. The scikit-learn MLP was trained for 120 iterations and tuned using different hidden layer sizes and activation functions in order to identify an effective architecture using the same RandomSearchCV.

Our custom PyTorch neural network on the other hand was trained for 60 epochs using mini-batch gradient descent with the Adam optimizer. Hyperparameters such as learning rate, number of layers, hidden units, and dropout rates were optimized using Optuna across 30 trials and early stopping was employed in order to prevent overfitting by monitoring the model's validation performance.

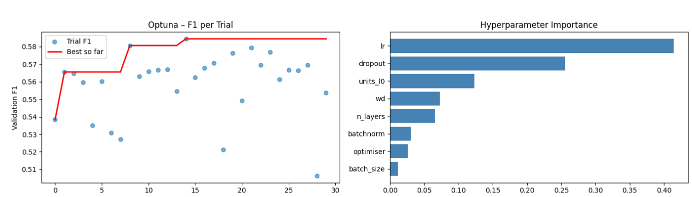


Fig. 2: Optuna Trials for Our Custom PyTorch Neural Network, ShotNet

To evaluate robustness of models across different historical periods and investigate recency bias, we employed a

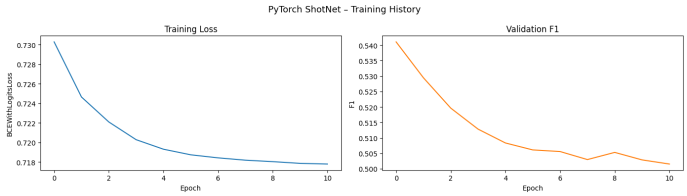


Fig. 3: Training Curves for Our Custom PyTorch Neural Network, ShotNet

separate set of temporal experiments. For these experiments, LightGBM models were trained with three distinct scenarios: rolling windows, expanding windows, and a head to head. Rolling windows consisted of fixed 3 season training intervals that moved forward in time. Expanding windows consisted of incrementally increasing training data either in the forwards or backwards direction. The head to head consisted of training on the oldest 30% of data and comparing that to the performance on the newest 30% of data. These models were primarily evaluated with F1-score, however the head to head also included evaluation of ROC-AUC to provide a threshold independent measure of model discrimination. Further implementation details for all experiments, exact hyperparameter ranges, and code are provided in Appendix A.

## V. Results

Prior to model training, the integration of distinct machine learning architectures was predicated on specific theoretical expectations regarding the nature of NBA shot data. The Logistic Regression model was deployed under the assumption that it would successfully capture linear spatial realities, most notably that shot probability decreases as distance from the basket increases. However, given the fast-paced, multi-variable nature of basketball, it was anticipated that this linear baseline would plateau early. Advanced tree-based ensembles (Random Forest and XGBoost) and neural architectures (scikit-learn's MLP and a custom PyTorch Feed Forward Neural Network) were subsequently implemented with the expectation that they would map the complex, non-linear interactions between spatial coordinates, time remaining, and generalized player biometrics (e.g., how the height and age of a player might interact with their optimal shooting zones). Furthermore, regarding the temporal analysis, the prevailing hypothesis was that training models on historically distant data (e.g., the mid-2000s) would negatively impact modern predictions due to the strategic paradigm shift of the "three-point revolution," suggesting that older data might act as noisy or outmoded context for evaluating modern shot efficiency.

The empirical results established a clear, albeit challenging, reality regarding the predictability of basketball shots. The Logistic Regression baseline, configured with balanced class weights, achieved a test accuracy of approximately 59.0% and a macro-average F1-score of 0.59. While this performance securely outpaces a random classifier, it underscores the inherent stochasticity of the sport. Even when controlling for precise spatial coordinates and player physicality, a basketball shot remains heavily influenced by unrecorded, micro-level physical events. The baseline served its primary purpose: it confirmed that spatial and biometric data carry predictive signal, but also highlighted the limitations of linear decision boundaries in this domain.

TABLE I: Logistic Regression Results on Test Set

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.62 | 0.62 | 0.62 | 22,946 |
| 1 | 0.55 | 0.56 | 0.56 | 19,365 |
| accuracy | — | — | 0.59 | 42,311 |
| macro avg | 0.59 | 0.59 | 0.59 | 42,311 |
| weighted avg | 0.59 | 0.59 | 0.59 | 42,311 |

To address these limitations, the study advanced to highly-tuned ensemble and neural network architectures. Among the scikit-learn and XGBoost models, the Random Forest classifier achieved the highest validation F1-score ( 0.511) and a test accuracy of 60.9%. The Multi-Layer Perceptron (MLP) demonstrated highly competitive capabilities, yielding a test accuracy of 61.4% (Val F1: 0.509). XGBoost, somewhat surprisingly for tabular data, slightly lagged behind the others in this specific hyperparameter configuration, returning a validation F1-score of 0.499. The relatively tight clustering of these advanced models, hovering around 60–61% accuracy, indicates that while tree ensembles and MLPs successfully capture additional non-linear variance, the marginal gain over the baseline is constrained by the hard ceiling of the dataset's inherent noise.

Ultimately, the custom PyTorch Feed Forward Neural Network (ShotNet), rigorously optimized via Optuna across 30 hyperparameter trials, yielded the highest raw performance of all tested models. The architecture achieved a test accuracy of 62.2% and a test F1-score of 0.508. While correctly classifying over 62% of the shots represents a notable analytical success in sports forecasting, the F1-score remained constrained in the low 0.50s. This limitation was primarily driven by lower recall on the positive class (made shots), suggesting that while the network is highly proficient at identifying shots with a high probability of missing, confidently isolating "guaranteed" made shots remains an elusive computational task.


Fig. 4: Comparison of Test-Set Performance Metrics Across the Advanced Machine Learning Architectures

TABLE II: Scikit-Learn and XGBoost Model Comparisons Sorted by Validation F1

| Model | Val Accuracy | Val Precision | Val Recall | Val F1 | Test Accuracy | Test Precision | Test Recall | Test F1 |
|---|---|---|---|---|---|---|---|---|
| Random Forest | 0.612512 | 0.604977 | 0.441931 | 0.510758 | 0.608622 | 0.598442 | 0.440279 | 0.507319 |
| Neural Network (MLP) | 0.614474 | 0.609986 | 0.437180 | 0.509325 | 0.613774 | 0.607511 | 0.441053 | 0.511070 |
| XGBoost | 0.609321 | 0.603869 | 0.425562 | 0.499273 | 0.605856 | 0.597194 | 0.426439 | 0.497575 |

A deeper examination of the models' confusion matrices reveals a pronounced discrepancy between precision and recall on the positive class (made shots), explaining the constrained F1-scores. The baseline Logistic Regression model, aided by balanced class weights, produced a relatively symmetric matrix on the validation set (14,143 True Negatives; 10,820 True Positives; 8,803 False Positives; 8,545 False Negatives).

However, as the models became more advanced, their prediction strategies became notably more conservative. The Random Forest test set evaluation heavily reduced False Positives (5,721) but saw False Negatives spike to 10,839. This trend peaked with the optimized PyTorch ShotNet, which correctly identified 18,019 missed shots (True Negatives) and only 8,282 made shots (True Positives), generating a massive 11,083 False Negatives. This high volume of False Negatives, where the model predicts a miss but the player actually makes the shot, highlights the boundaries of spatial data and occurs due to several overlapping factors. First, the architectures exhibit a threshold-driven conservatism where they prioritize mathematical probability thresholds over raw recall; given that mean jump-shot conversion rates typically fluctuate between 35-45%, the binary classifiers optimize for global accuracy by defaulting to a negative prediction for the majority of non-restricted area attempts. Second, the absence of high-fidelity context regarding the latent defender variable, such as defensive proximity and contest intensity, forces models to regress toward historical spatial means, fundamentally limiting their capacity to confidently assign a positive outcome to high-quality, open-look opportunities. Finally, this analytical ceiling is reinforced by the idea that top-tier NBA athletes frequently convert attempts characterized by extreme mathematical improbability. Since the unquantifiable nuances of human skill and defensive pressure are not captured within the feature set, the advanced models preserve predictive integrity by leaning heavily toward a negative classification, a strategy that prioritizes precision but results in a significant sacrifice of positive recall.

A defining objective of this research was to quantify temporal influence across the spanning 2004–2024 dataset, utilizing a LightGBM classifier evaluated against a fixed modern test set. AUC was selected as a primary metric for the temporal analysis as it provides a threshold-invariant measure of the model's discriminatory power, ensuring that the evaluation of historical models on modern data remains robust against era-specific shifts in baseline shooting percentages and class imbalances. The results of these experiments fundamentally challenged the team's initial hypothesis regarding recency bias. When evaluated using a rolling 3-season training window, model performance remained remarkably stable across the two-decade span, maintaining a Test AUC of approximately 0.630, with only minor deviations in F1-scores observed during the league's transitional years (2011–2016).

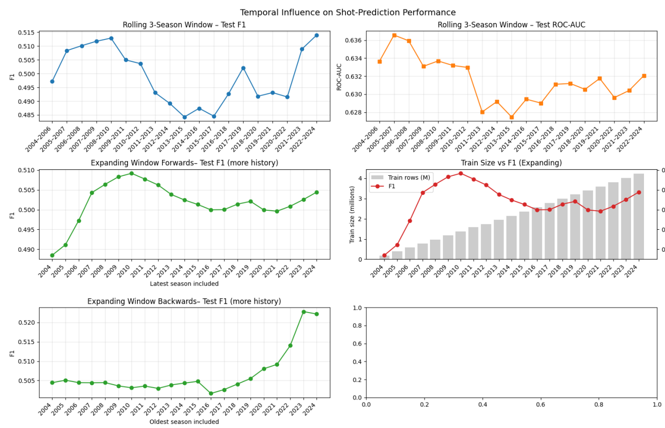


Fig. 5: Temporal Influence on Shot Prediction Performance

The expanding window experiment further dismantled the assumption that older data degrades modern predictive power. Progressively adding historical data to the training set demonstrated a slow but consistent improvement in model performance. Training exclusively on the 2004 season yielded an AUC of 0.630; however, expanding the training window cumulatively to include all seasons up to 2024 elevated the AUC to 0.640. This confirms that adding more data, regardless of the era it originates from, allows the model to better learn universal shooting fundamentals rather than confusing it with outdated stylistic trends.

This resilience against recency bias was definitively confirmed in the head-to-head evaluation. A direct comparison was drawn between a model trained exclusively on the oldest 30% of the dataset (the 6 seasons from 2004–2009) and one trained on the newest 30% (the 6 seasons from 2019–2024). Astonishingly, the early-era model predicted modern test outcomes with an accuracy of 62.2%, an F1 of 0.508, and an AUC of 0.638. The model trained on modern data performed virtually identically in accuracy ( 62.2%) but registered slightly lower secondary metrics (F1: 0.505, AUC: 0.632). These results lead to the conclusion that while the volume, value, and strategy of specific shots (such as the high-volume three-pointer) have radically evolved, the fundamental laws that dictate whether a physical basketball goes through a physical hoop remain timeless.

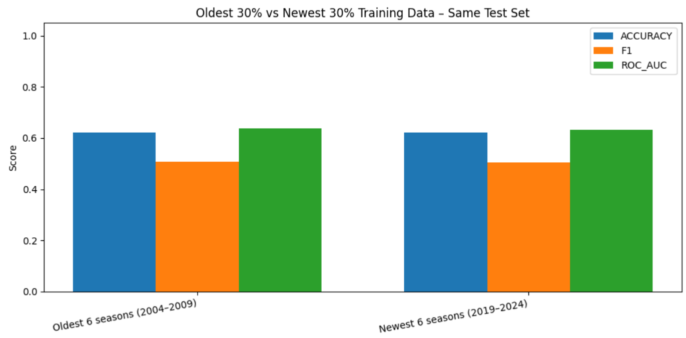


Fig. 6: Head-to-Head Evaluation Using Oldest Versus Newest Data

## VI. Discussion

Contextualizing these findings with respect to the related literature provides insights into the evolution of basketball modeling. The predictive ceiling of 62.2% achieved by the Optuna-tuned PyTorch ShotNet aligns closely with, but slightly trails, the 64-68% accuracies reported in foundational studies by Meehan [1] and Alves et al. [6]. This variance is instructive; while prior works often localized their training to single seasons (e.g., the 2014-15 dataset) or retained specific player identities, our models were tasked with generalizing across two decades using only anonymized spatial and biometric features. The convergence of our advanced models around the low-60s percentile empirically validates the "stochastic ceiling" proposed by Alves et al. [6], proving that across a massive, longitudinal dataset, the inherent physical unpredictability of the sport severely bounds purely spatial predictive power.

Furthermore, the pronounced precision-recall anomaly observed in our models (specifically the generation of 11,083 False Negatives by the PyTorch architecture) serves as a direct validation of the conclusions drawn by Harmon et al. [3] and Sliz [4]: our dataset lacks the high-fidelity optical tracking data (SportVU) utilized in those studies; our models were blind to defensive proximity and contest intensity. The extreme conservatism of our tree ensembles and neural networks demonstrates exactly what happens when predictive algorithms lack this dynamic context: unable to distinguish an open look from a heavily contested shot, the models are forced to regress to historical spatial averages, routinely misclassifying mathematically improbable shots converted by elite athletes.

Finally, our temporal findings provide a necessary expansion to the longitudinal trends documented by Zajac et al. [5]. While Zajac et al. thoroughly mapped the strategic transformation of the league during the "three-point revolution," our rolling and expanding window experiments prove that this shift was entirely compositional, not mechanical. By demonstrating that a model trained on 2004-2009 data predicts 2019-2024 outcomes with near-identical accuracy( 62.2%), we establish that while the strategic volume and location of shot attempts have radically changed as noted in prior literature, the underlying physical probabilities governing the interaction between player biometrics, distance, and the hoop have remained strictly invariant over the last twenty years.

## VII. Conclusion

This research successfully developed a strong machine learning pipeline capable of predicting binary NBA shot outcomes utilizing two decades of spatial and generalized biometric data. Through the rigorous evaluation of linear baselines, tree-based ensembles, and a custom, Optuna-tuned PyTorch Feed Forward Neural Network (ShotNet), the study identified a predictive accuracy ceiling of approximately 62.2%. Crucially, the project illuminated a pronounced precision-recall discrepancy, revealing that models inherently adopt conservative, threshold-driven prediction strategies when faced with stochastic human actions. Furthermore, the temporal analysis yielded a pivotal contribution to sports analytics by empirically disproving the assumption of recency bias. Models trained on early-2000s data generalized seamlessly to modern test sets, demonstrating that while the strategic volume of specific shots has evolved, the foundational mechanical and spatial laws governing shot success remain timeless.

Beyond sports analytics, these findings present profound implications for the broader field of artificial intelligence, particularly regarding dataset curation and contextual limitations. The precision-recall anomaly underscores a fundamental constraint of AI when forecasting stochastic events: algorithms optimize for global mathematical probability and struggle to account for unquantifiable human exceptionalism or missing latent variables (such as defensive proximity). Additionally, the temporal resilience of the early-era models serves as a vital case study in machine learning. It suggests that for AI systems grounded in underlying physical, mechanical, or biological laws, historical data does not inherently degrade over time. This challenges a prevailing industry assumption that older data acts as "noise," proving that sheer data volume spanning long temporal horizons can reinforce core fundamentals better than restricted, ultra-modern datasets.

To transcend the current predictive ceiling of 62%, future research on this topic must prioritize contextual feature completeness over pure algorithmic complexity. The most critical improvement involves the integration of high-fidelity optical tracking data to quantify real-time defender proximity and contest intensity, directly addressing the contextual void responsible for the models' high volume of false negatives. Furthermore, rather than treating each shot as an isolated spatial event, future work should explore sequence-based deep learning architectures, such as Long Short-Term Memory networks or Transformers. By feeding the model the sequential progression of player movements, passes, and momentum leading up to the release of the ball, predictive systems can begin to quantify the multi-variable reality of professional basketball.

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## Appendix A
## Code and Data Availability

For full reproducibility, the complete codebase, data processing pipeline, feature engineering, model training, hyperparameter optimization, and temporal analysis for this project are publicly available in the following resources:

- **GitHub Repository** (main project repository containing all notebooks, scripts, and processed data): https://github.com/RyanStyron/NBAShotPredictor/tree/main
- **Main Shot Prediction Pipeline** (Colab notebook covering data retrieval, merging, EDA, modeling, and LightGBM temporal analysis): https://colab.research.google.com/drive/1Z2egQVk9wMvgxuJdX1dQfK9hZb7uheMp?usp=sharing
- **NBA Player Biometric Data Retrieval Notebook** (detailed `nba_api` pipeline used to construct the biometric dataset): https://drive.google.com/file/d/10f5eky55QIbNHKonSbDR38sJq7vyYU8w/view?usp=sharing

The datasets used in this study are publicly hosted on Kaggle and formally cited in the References section:

- NBA Shots (2004–2024) [7]
- Biometric Attributes of NBA Players per Season [8]

Player biometric data was retrieved using the open-source `nba_api` Python library (https://github.com/swar/nba_api).

## Appendix B
## Team Member Contributions

This project was a collaborative effort among all four team members. The specific contributions of each author toward the completion of the final project are as follows:

- **Harman Aujla**: data preprocessing, feature engineering, baseline model, shot examples
  - Helped engineer the final dataset by dropping features with high VIF and any NaNs
  - Split the dataset into train, val, and test sets and ran a baseline logistic regression model
  - Using the best model, illustrated examples of correctly and incorrectly classified results
- **Matthew Voynovich**: ensemble and neural models, temporal analysis, model tuning
  - Developed and implemented all ensemble models (Random Forest, XGBoost) and neural network models (scikit-learn MLP and custom PyTorch network) for shot outcome prediction
  - Designed and executed hyperparameter tuning pipelines across all models using RandomizedSearchCV and Optuna to optimize performance
  - Conducted temporal analysis using LightGBM models, implementing rolling and expanding window experiments to evaluate model stability across seasons
  - Generated and visualized model results, including performance comparisons, confusion matrices, and training curves for neural networks
- **Ryan Styron**: data preprocessing, feature engineering, analysis of results
  - Engineered the final modeling dataset by identifying and removing multicollinear spatial features, rare action types, and target-derived variables to prevent data leakage.
  - Executed the preprocessing pipeline by applying one-hot encoding to categorical game-context variables and standardizing numerical player biometrics for model compatibility.
  - Conducted the comparative performance analysis between the logistic baseline, advanced tree ensembles, and feed-forward neural networks, specifically diagnosing the precision-recall anomaly and interpreting the threshold-driven conservatism of the models.
  - Evaluated the custom PyTorch ShotNet, analyzing the Optuna hyperparameter tuning trials to formally establish the project's 62.2% predictive accuracy ceiling.
- **Joseph Fodera**: data retrieval and aggregation, literature review
  - Researched related works to help form the basis of our project and define gaps we were looking to fill.
  - Searched for viable datasets.
  - Created 20 unique biometric datasets respective to each season from NBA API.
  - Merged and anonymized a total of 40 datasets into one, dropping columns deemed unnecessary to curate the dataset used for the project.

All team members participated in project planning, literature review, experimental design discussions, result interpretation, and the writing and revision of the final report.